CLOUD SECURITY SPECIALIST

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LAB:SECURING AZURE SQL DATABASE

1.1 INTRODUCTION

I started the lab with a required lab set up.I clicked on ctrl alt and delete to bring up the login page,signed in as admin and then downloaded the lab files.

1.2 PREPARE CLOUD SHELL FOR LATER USE

Step 1

I logged in to the Azure portal using the credentials provided in the lab

Step 2

In the toolbar at the top of the Azure portal, I selected the Cloud Shell icon

Step 3

In the Welcome to Azure Cloud Shell dialog, I selected PowerShell

Step 4

On the Getting Started screen,I selected Mount Storage Account.

Step 5

In the Storage account subscription drop down, I selected my subscription then clicked the apply button.

Step 6

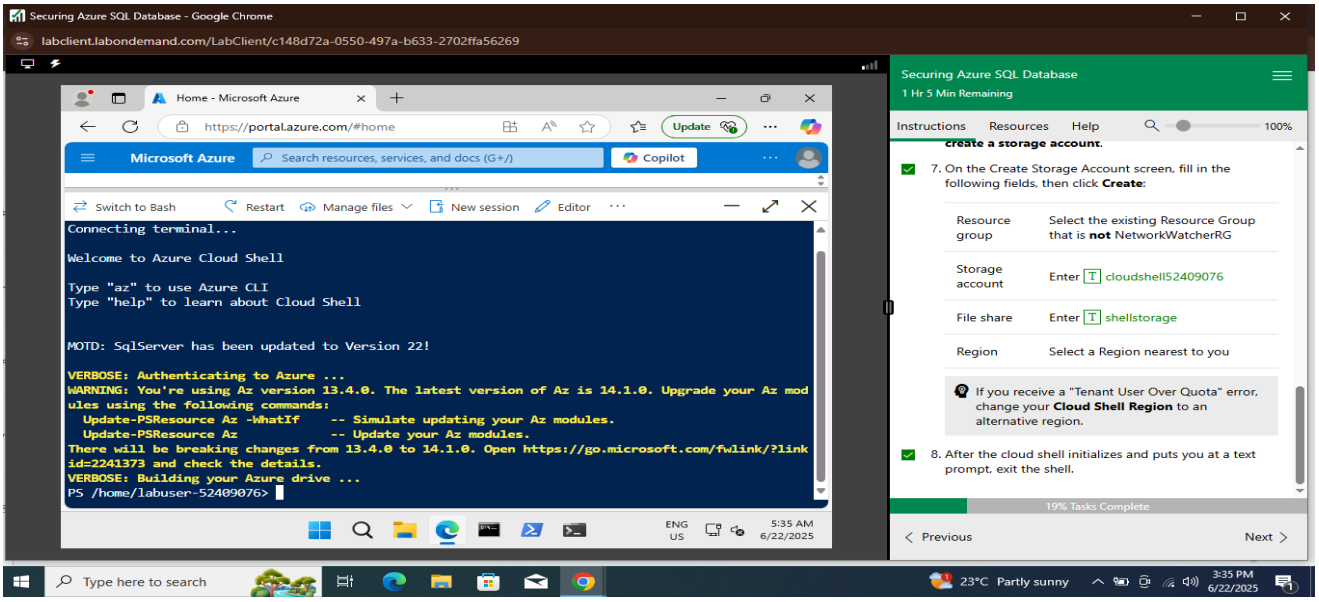
In the Mount Storage Account screen, I selected the I want to create a storage account

Step 7

On the Create Storage Account screen, I filled in the fields for;resource group,storage account,file share and region then clicked on the Create button.

Step 8

The cloud shell initialized and put me at a text prompt, I then exited the shell. Here is my screenshot



1.3 LAB 05:SECURING AZURE SQL DATABASE

LAB SCENARIO

In this lab,I reviewed security features for Azure SQL databases specifically with interest in:Protection against attacks such as SQL injection and data exfiltration,ability to discover and classify database information into categories such as Confidential and ability to audit database server and database queries and log events.

EXERCISE 1:IMPLEMENT SQL DATABASE SECURITY FEATURES

In this exercise,I completed the tasks;Deploy an Azure SQL Database,Configure Advanced Data Protection,Configure Data Classification and Configure Auditing.

TASK 1:DEPLOY AN AZURE SQL DATABASE

In this task,I used a template to deploy the lab infrastructure.

Step 1

I signed in to the Azure portal

Step 2

In the Azure portal, in the Search resources, services, and docs text box at the top of the Azure portal page, I typed Deploy a custom template and pressed the Enter key

Step 3

On the Custom deployment blade, I clicked the Build your own template in the editor option

Step 4

On the Edit template blade, I clicked Load file, located the \Allfiles\Labs\11\azuredeploy.json file and clicked Open

Step 5

On the Edit template blade, I clicked Save

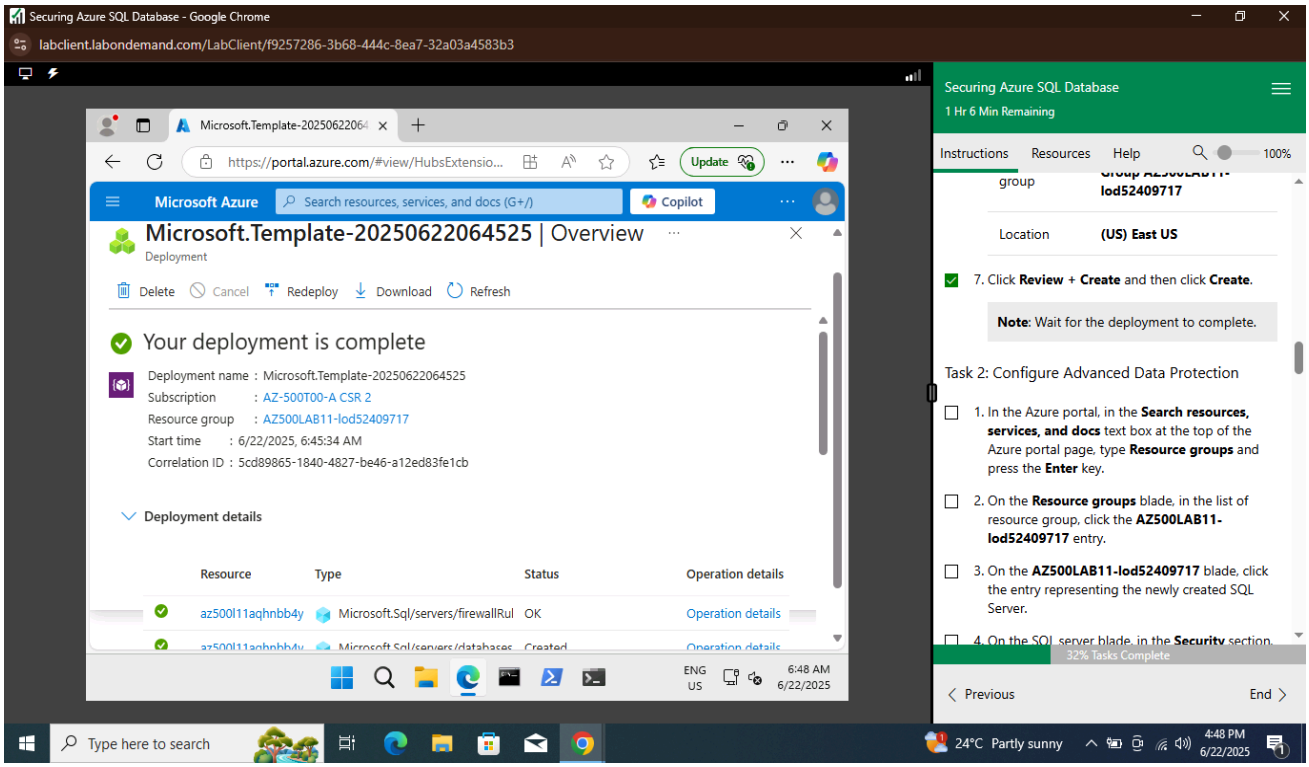
Step 6

On the Custom deployment blade, I ensured that the settings for;subscription,resource group,and location were configured and left any others with their default values

Step 7

I clicked on Review + Create and then Create buttons.

Below is my screenshot



TASK 2:CONFIGURE ADVANCED DATA PROTECTION

Step 1

In the Azure portal, in the Search resources, services, and docs text box at the top of the Azure portal page, I typed Resource groups and pressed the Enter key

Step 2

On the Resource groups blade, in the list of resource group, I clicked the AZ500LAB11-lod52409717 entry

Step 3

On the AZ500LAB11-lod52409717 blade, I clicked the entry representing the newly created SQL Server.

Step 4

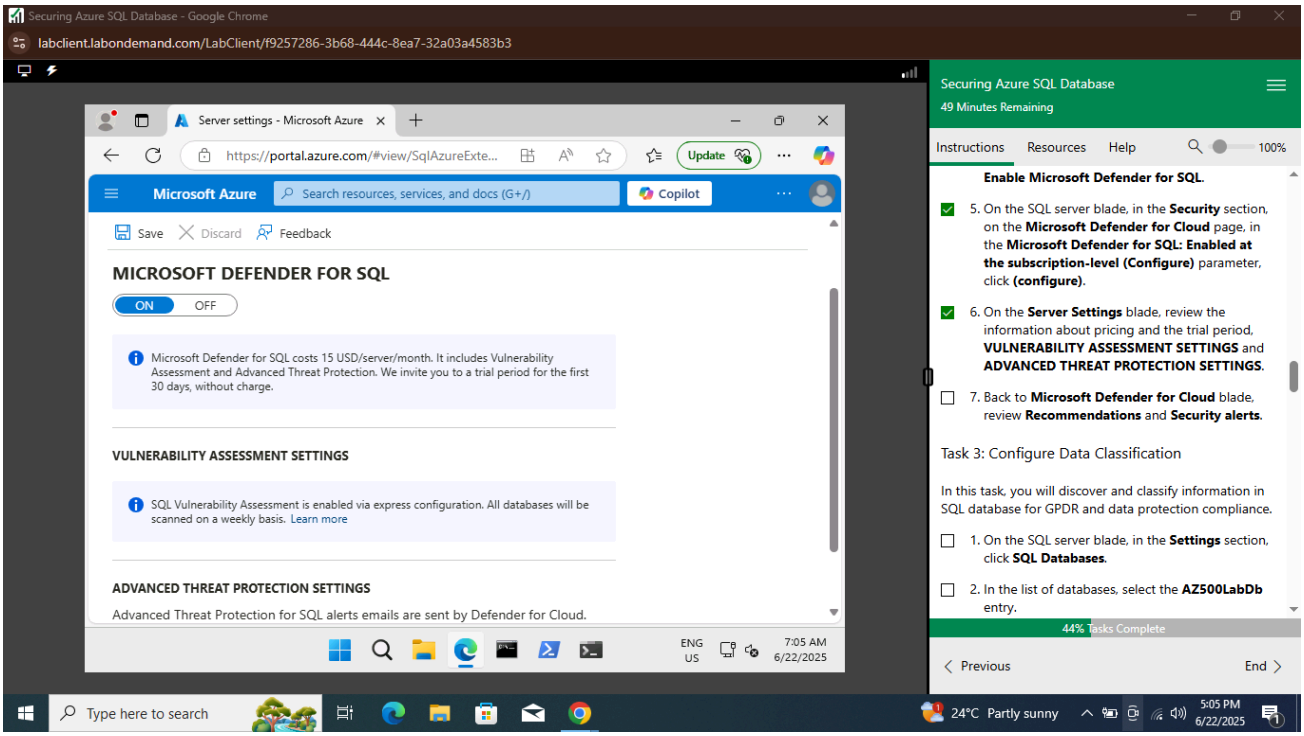
On the SQL server blade, in the Security section, I clicked Microsoft Defender for Cloud, and selected Enable Microsoft Defender for SQL

Step 5

On the SQL server blade, in the Security section, on the Microsoft Defender for Cloud page, in the Microsoft Defender for SQL: Enabled at the subscription-level (Configure) parameter, I clicked on (configure)

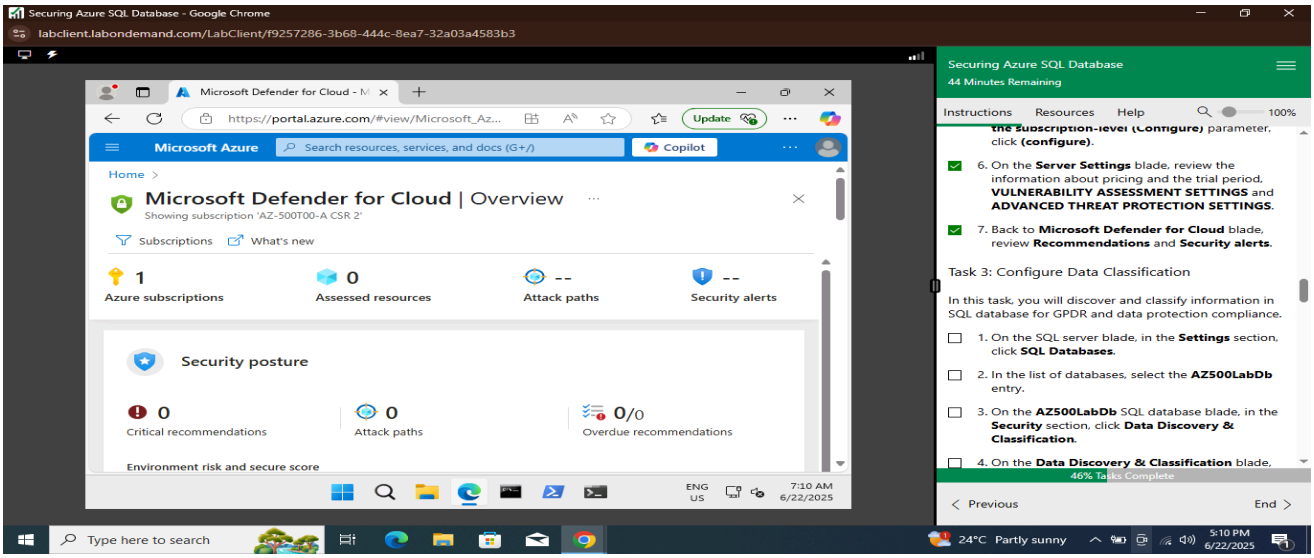
Step 6

On the Server Settings blade, I reviewed the information about pricing and the trial period, VULNERABILITY ASSESSMENT SETTINGS and ADVANCED THREAT PROTECTION SETTINGS.Here is my screenshot



Step 7

Back to Microsoft Defender for Cloud blade,I reviewed Recommendations and Security alerts.Here is my screenshot



TASK 3:CONFIGURE DATA CLASSIFICATION

In this task,I discovered and classified information in the SQL database for GPDR and data protection compliance.

Step 1

On the SQL server blade, in the Settings section, I clicked on SQL Databases

Step 2

In the list of databases, I selected the AZ500LabDb entry

Step 3

On the AZ500LabDb SQL database blade, in the Security section, I clicked on Data Discovery & Classification

Step 4

On the Data Discovery & Classification blade, I clicked the Classification tab

Step 5

I clicked the text message We have found 15 columns with classification recommendations displayed on blue bar at the top of the blade

Step 6

I reviewed the listed columns and the recommended sensitivity label

Step 7

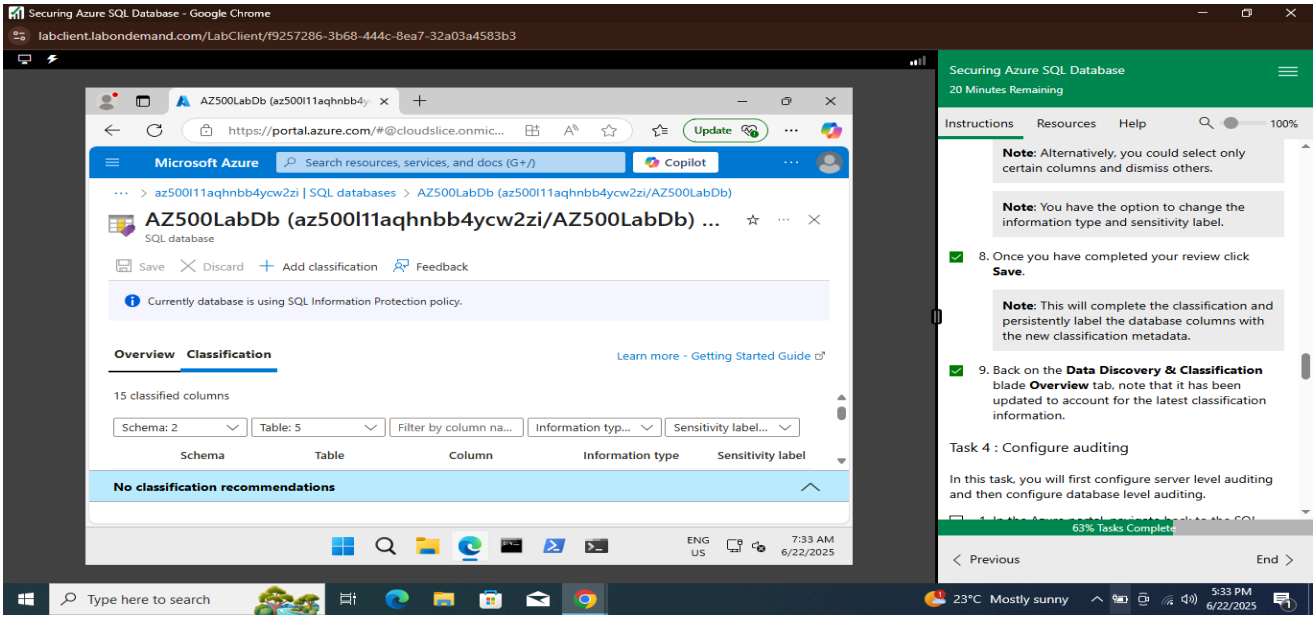
I then enabled the Select all checkbox and clicked Accept Selected Recommendations.

Step 8

Once I completed my review, I clicked on Save

Step 9

Back on the Data Discovery & Classification blade Overview tab,I noted that it has been updated to account for the latest classification information.Here is my screenshot



TASK 4:CONFIGURE AUDITING

In this task,I first configured server level auditing and then configured database level auditing

Step 1

In the Azure portal, I navigated back to the SQL Server blade

Step 2

On the SQL Server blade, in the Security section, I clicked on Auditing

Step 3

I set the Enable Azure SQL Auditing switch to ON to enable auditing

Step 4

I selected the Storage checkbox and entry boxes for Subscription and Storage Account were displayed

Step 5

I chose my subscription from the dropdown list

Step 6

I clicked the storage account and chose Create new

Step 7

On the create storage account blade, in the Name box, I typed a globally unique name consisting of between 3 and 24 lower case letters and digits, and clicked the OK button

Step 8

Back on the Auditing blade, under Advanced properties, I set Retention (days) to 5 days

Step 9

On the Auditing blade, I clicked save to save the auditing settings

Step 10

On the server blade, in the Settings section, I clicked on SQL Databases

Step 11

In the list of databases, I selected the AZ500LabDb entry

Step 12

On the AZ500LabDb SQL database blade, in the Security section, clicked Auditing

Step 13

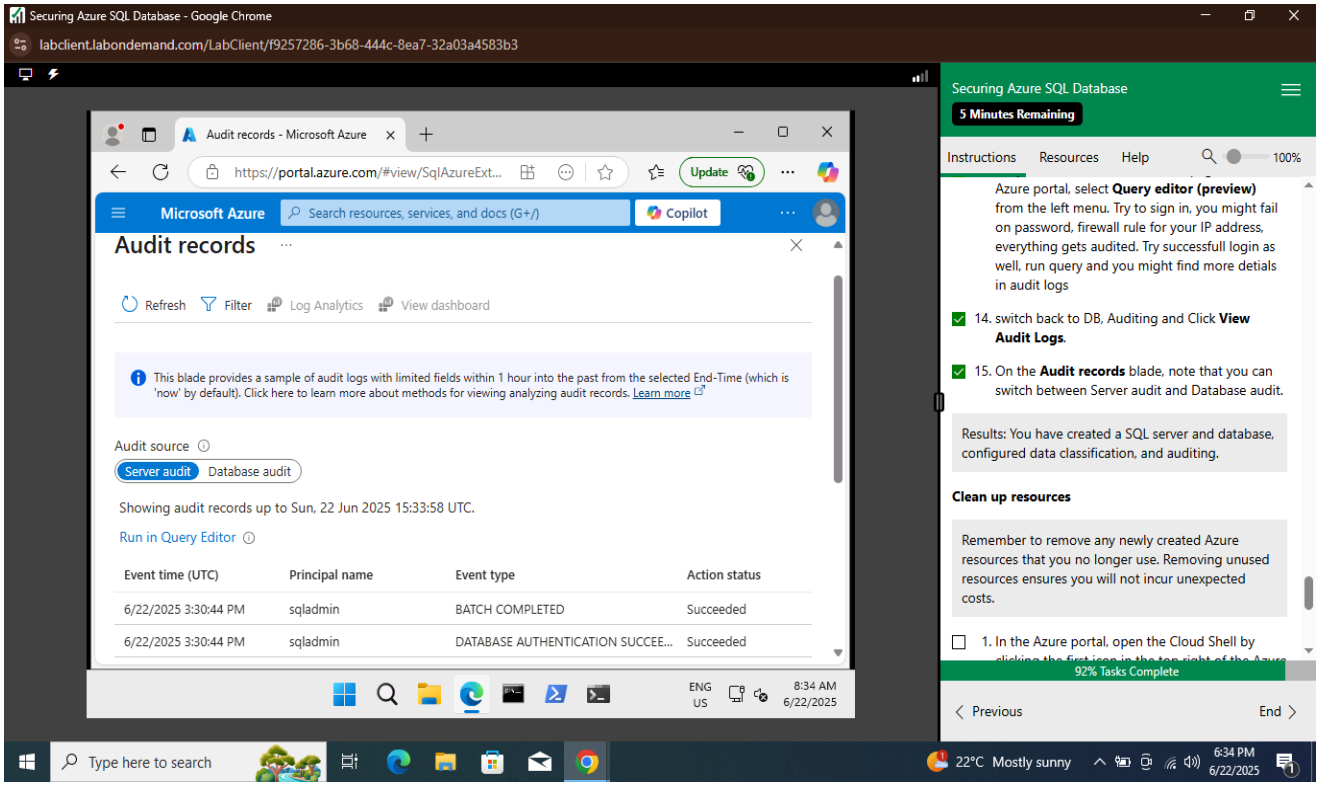
On my SQL database Overview page in the Azure portal, I selected the Query editor (preview) from the left menu and signed in.

Step 14

I switched back to DB, Auditing and clicked on view Audit Logs

Step 15

On the Audit records blade, I noted that I can switch between Server audit and Database audit. Below is my screenshot



I cleaned up my resources and completed the lab